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Light-Responsive Enzyme-Loaded Nanoparticles for Tunable Adhesion and Mechanical Wound Contraction

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Tissue adhesives have emerged as minimally invasive tools for wound closure and the secure attachment of biomaterials to tissue surfaces. However, the development of on-demand, tunable adhesives remains limited, restricting their adaptability to dynamic and complex tissue interfaces. In this study, a spatiotemporally light-responsive and synergistic system: photoactivatable tyrosinase-loaded mesoporous polydopamine nanoparticles (MPDA_PaTy) is presented. Upon ultraviolet (UV) irradiation, MPDA_PaTy enables precise modulation of adhesive strength, achieving up to 3.7-fold enhancement with increasing irradiation time. This tunability facilitates effective tissue adhesion for in vivo wound closure in mouse incision models and enhances the mechanical contraction of circular wounds by anchoring pre-stretched hydrogels, as demonstrated in both ex vivo and in vivo models. Finite-element modeling further confirms the hydrogel's wound-contraction capability. These results demonstrate that MPDA_PaTy is a versatile, tunable bioadhesive with significant potential for advanced tissue engineering applications.

1. Introduction

Tissue adhesives offer robust performance in various surgical and clinical fields, including wound sealing and mechanical support to tissues or patches.[1,2] For instance, hydrogels and nanoparticles have been developed as non-invasive adhesives for tissue regeneration and wound closure. These approaches enable faster and less painful healing, reducing complications compared to invasive methods such as sutures or staples. However, they remain slow, passive, and limited in their tunability of adhesive forces, which restricts their broader application and effectiveness.[3] This highlights the need for a dynamic, on-demand, tunable adhesion system that can be delicately adjusted to enable moderate adhesion for easy repositioning, or strengthened for tissue reinforcement.

Nanoparticles provide a large surface area capable of adsorbing numerous monomers with an affinity for their surface, effectively acting as connectors.[4–6] When absorbed onto hydrogel or tissue surfaces, these nanoparticles facilitate interactions between polymer chains in hydrogels and protein chains in biological tissues.[7] This property has shown potential for wound closure by enhancing connections at wound sites.[4–6,8] In this study,

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mesoporous polydopamine nanoparticles (MPDA), with a large loading space and intrinsic adhesive properties derived from catechol and quinone groups, were used as tissue adhesives. Catechol interacts noncovalently with functional groups through hydrogen bonding, electrostatic interactions, and π–π stacking.[9,10] Upon oxidation to quinone, catechol can also form covalent bonds with tissue surface groups, such as amines, thiols, and imidazoles, via Schiff-base reactions and Michael-type addition reactions. Porous carriers, such as mesoporous silica nanoparticles, have been explored for drug delivery[11,12] and tissue adhesion[8,13,14] due to their large loading capacity and high surface area. Thus, MPDA combines a mesoporous structure with inherent catechol-based adhesion, making it particularly suited for tissue- and hydrogel-adhesive applications.

Tyrosinase, an enzyme that oxidizes phenols to catechols and quinones, facilitates covalent bonds with primary amines, thiols, or imidazoles through enzymatic reactions under physiological conditions.[15] This property has been used to crosslink polymer chains in tissue-like hydrogels and protein chains within biological tissues, demonstrating the potential for wound healing and tissue engineering.[16–18] Our group demonstrated that tyrosinase enhances tissue adhesion by crosslinking protein chains within tissues,[19] resulting in increased collagen density and improved wound closure.[20]

We have previously reported an enzymatic polymer crosslinking system using *Streptomyces avermitilis*-derived recombinant tyrosinase (SaTy), which exhibits higher crosslinking efficiency compared to other tyrosinases due to its wide substrate entrance and the short distance from the enzyme surface to its active site.[17,21–23] Since the I41Y mutant tyrosinase exhibited activity similar to the wild type, we substituted the residue with *o*-nitrobenzyl tyrosine (ONBY), a bulky and photocleavable analog, thereby successfully generating a photoactivatable variant.[24,25] The ONBY group disrupts substrate binding, making the enzyme inactive. Ultraviolet (UV) exposure cleaves the ONBY group, restoring enzymatic activity and enabling on-demand activation (**Figure 1A**). While previously demonstrated for melanin biosynthesis and simple hydrogel formation, photoactivatable tyrosinase (PaTy) in this study is repurposed for tunable, synergistic tissue adhesion and wound contraction.

Accordingly, we constructed a tunable adhesive system by integrating PaTy with MPDA for both tissue-tissue and tissue-hydrogel interfaces. PaTy-loaded MPDA (MPDA_PaTy) first enables immediate adhesion via adsorption and non-covalent interactions. Second, UV irradiation activates PaTy to induce enzymatic crosslinking with polymer networks or extracellular matrix (ECM) components, thereby enhancing adhesive strength. This approach induces phenol-specific oxidation, allowing precise on-demand crosslinking with minimal inflammation.[1] Additionally, MPDA interacts with tyrosinase-activated polymers or ECM to reinforce bonding synergistically.

Figure 1 illustrates the schematic strategies involving PaTy activation and its application in two distinct interfaces. UV exposure triggers the removal of the ONBY group, thereby activating PaTy and enabling enzymatic crosslinking for adhesion (Figure 1A). We demonstrated on-demand enhancement of adhesive strength following UV irradiation of MPDA_PaTy at the tissue-tissue interface in in vivo mouse incision models. Additionally, we developed a stretchable hydrogel preloaded in a stretched state to achieve mechanical contraction while maintaining adhesion at the tissue-hydrogel interface, supported by MPDA_PaTy (Figure 1B). By combining the immediate adhesion capability of MPDA with enzyme-triggered covalent bonding by PaTy, we established a tunable adhesion platform. This synergistic MPDA_PaTy system not only provides precise control over adhesion but also enhances mechanical wound closure and promotes tissue regeneration, as demonstrated in both ex vivo porcine tissues and in vivo mouse models.

2. Results

2.1. Preparation and Characterization of Photoactivatable Tyrosinase-Loaded Mesoporous Polydopamine Nanoparticle (MPDA_PaTy)

To develop a tunable and on-demand tissue adhesive system, we engineered MPDA_PaTy by loading PaTy into MPDA, enabling controlled enzymatic tissue crosslinking upon UV activation (**Figure 2A**). MPDA was fabricated with an optimal size and porous structure, providing sufficient internal space to load and deliver PaTy effectively (Figure 2). Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images demonstrate the mesoporous structure of the MPDA (Figure 2B, Figure S1, Supporting information). The hydrodynamic size of MPDA was 103.1 ± 2.15 nm, and the zeta potential was −28.3 ± 1.5 mV (Figure 2C). For a synergistic and tunable adhesion system, unactivated PaTy was loaded on MPDA. As the PaTy loaded on the MPDA at mass ratios of 0.1:1, 0.25:1, 0.5:1, 1:1, and 2:1, the loading amount of PaTy has increased from 0.1 to 3.2 ± 0.2 U mL^{-1} (Figure S2, Supporting information). As the loading amount of PaTy hardly affected the colloidal stability of MPDA and reached saturation, a PaTy: MPDA ratio of 2:1 was used in all subsequent experiments with MPDA_PaTy (Figure S2A, Supporting information).

Loading of PaTy was also confirmed by Fourier transform infrared (FT-IR) spectra. The peak at 1635 cm^{-1} corresponds to C═O stretching (amide I), while the peak at 1564 cm^{-1} corresponds to N—H bending and C—N stretching (amide II). In addition, the peak at 2977 cm^{-1} is attributed to C—H vibrations, further confirming the presence of PaTy.[26] Attenuation of the broad O—H/N—H stretching band at ≈3300 cm^{-1} suggests hydrogen bonding and other non-covalent interactions between PaTy and the MPDA surface, providing evidence for PaTy loading (Figure S3, Supporting information).[27] The release behavior of MPDA_PaTy further confirmed PaTy loading, with complete release observed within 30 min (Figure S4, Supporting information).

SEM images of MPDA_PaTy were observed (Figure 2D). The MPDA_PaTy had a zeta potential of −26.8 ± 0.8 mV and a hydrodynamic size of 102.2 ± 3.7 nm, which indicates PaTy-loading didn't affect the size and the zeta potential (Figure 2E). MPDA_PaTy exhibited comparable enzymatic activity to free PaTy, with an 83.9 ± 2.5% efficiency of free PaTy (Figure 2F). Moreover, MPDA_PaTy retained 95% of its initial enzymatic activity for 7 days when stored at 4 and 25 °C (Figure S5, Supporting information).

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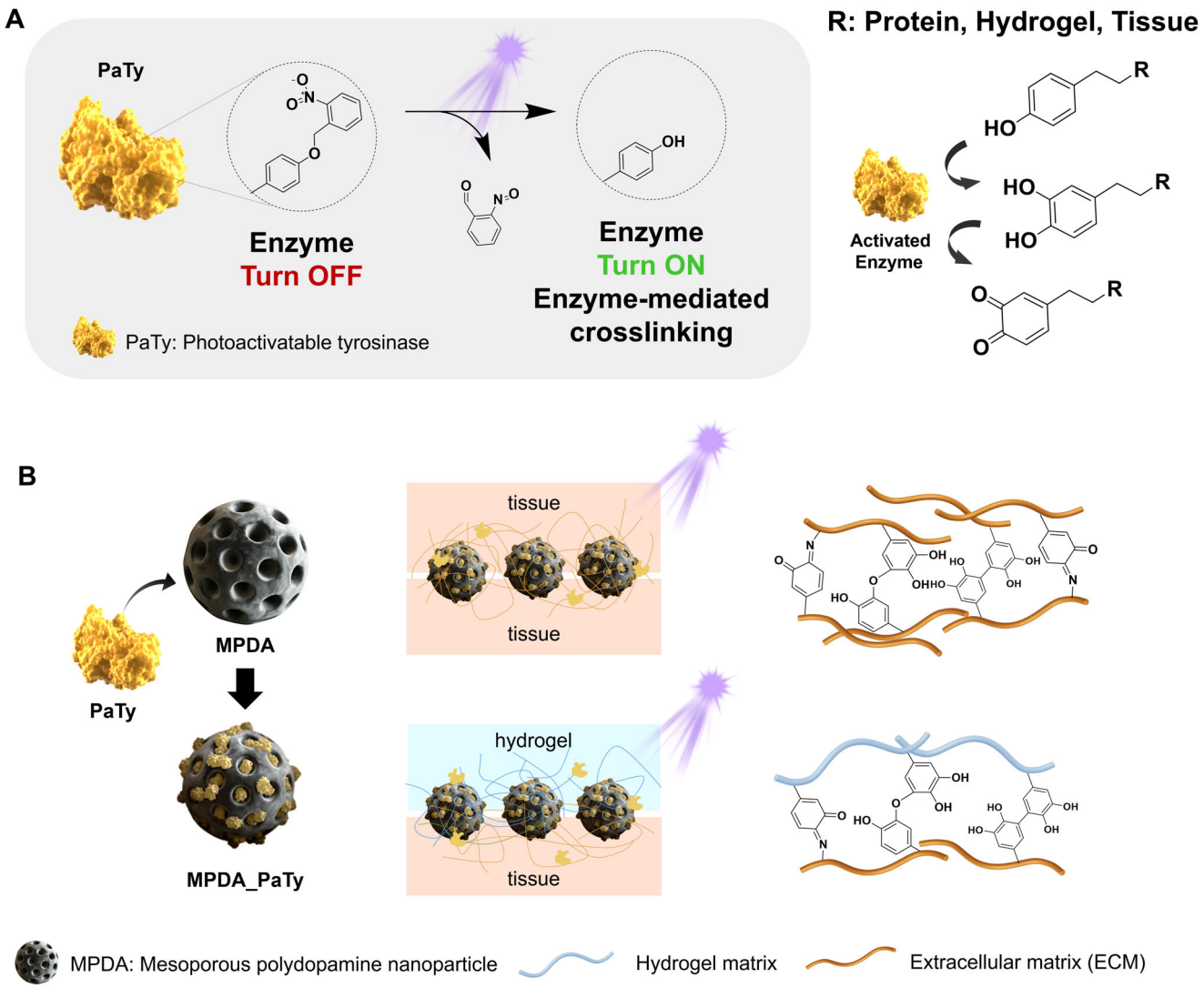

Figure 1. A) Schematic illustration of photoactivatable tyrosinase (PaTy) activation via ONBY removal and the subsequent oxidation of phenol and catechol residues by PaTy. B) Schematic illustration of PaTy-loaded mesoporous polydopamine nanoparticle (MPDA_PaTy) and the crosslinking of tissue-tissue and tissue-hydrogel interfaces by light-activated MPDA_PaTy. Activated catechol and quinone groups undergo coupling reactions or react with amine or thiol groups via Schiff-base reactions and Michael additions.

2.2. Tissue-Tissue Adhesion Properties of MPDA and MPDA_PaTy

Synergistic adhesion by MPDA_PaTy for tissue-tissue interfaces was demonstrated on ex vivo porcine skin through a lap-shear test (ASTM F2255) (Figure 2G–I). To determine the optimal concentration of MPDA solution for tissue adhesion, we compared adhesive strengths at different concentrations (1, 2, 3, and 4 mg mL^{-1}) (Figure 2G). The control group showed an adhesive strength of 0.4 ± 0.1 kPa. The adhesive strength increased to 1.0 ± 0.4 kPa with 1 mg mL^{-1} MPDA solution and peaked at 1.9 ± 0.5 kPa with 2 mg mL^{-1}. Higher concentrations, however, led to reduced strengths of 1.5 ± 0.4 kPa (3 mg mL^{-1}) and 1.1 ± 0.1 kPa (4 mg mL^{-1}). The adhesive strength of MPDA_PaTy was 1.3 ± 0.6 kPa before activation, and after photo-activation, the adhesive strength increased to 4.6 ± 1.2 kPa (Figure 2H). The loaded PaTy: MPDA ratio of 2:1 showed a significantly higher adhesive strength (4.6 ± 1.2 kPa) compared to MPDA with lower amounts of PaTy (2.7 ± 0.8 kPa for 1:1 and 1.4 ± 0.9 kPa for MPDA alone), confirming it as the optimal ratio (Figure S2C, Supporting information). Moreover, the tunable adhesive strength was demonstrated by varying the light irradiation on MPDA_PaTy. As the time of the irradiation increased, the adhesive strength between tissues was increased by activated PaTy, from 1.7 ± 0.5 kPa at 1 min irradiation to 2.7 ± 1 kPa at 3 min irradiation, reaching a 265% increase compared to the pre-activation state after 5 min (Figure 2I).

The FT-IR spectrum of collagen, a major component of the ECM, with activated MPDA_PaTy, demonstrated ECM crosslinking induced by PaTy activation. Peaks at 3290, 1554, and 1045 cm^{-1} showed an increase, indicating an increase in phenolic O—H groups due to the oxidation of phenolic groups into

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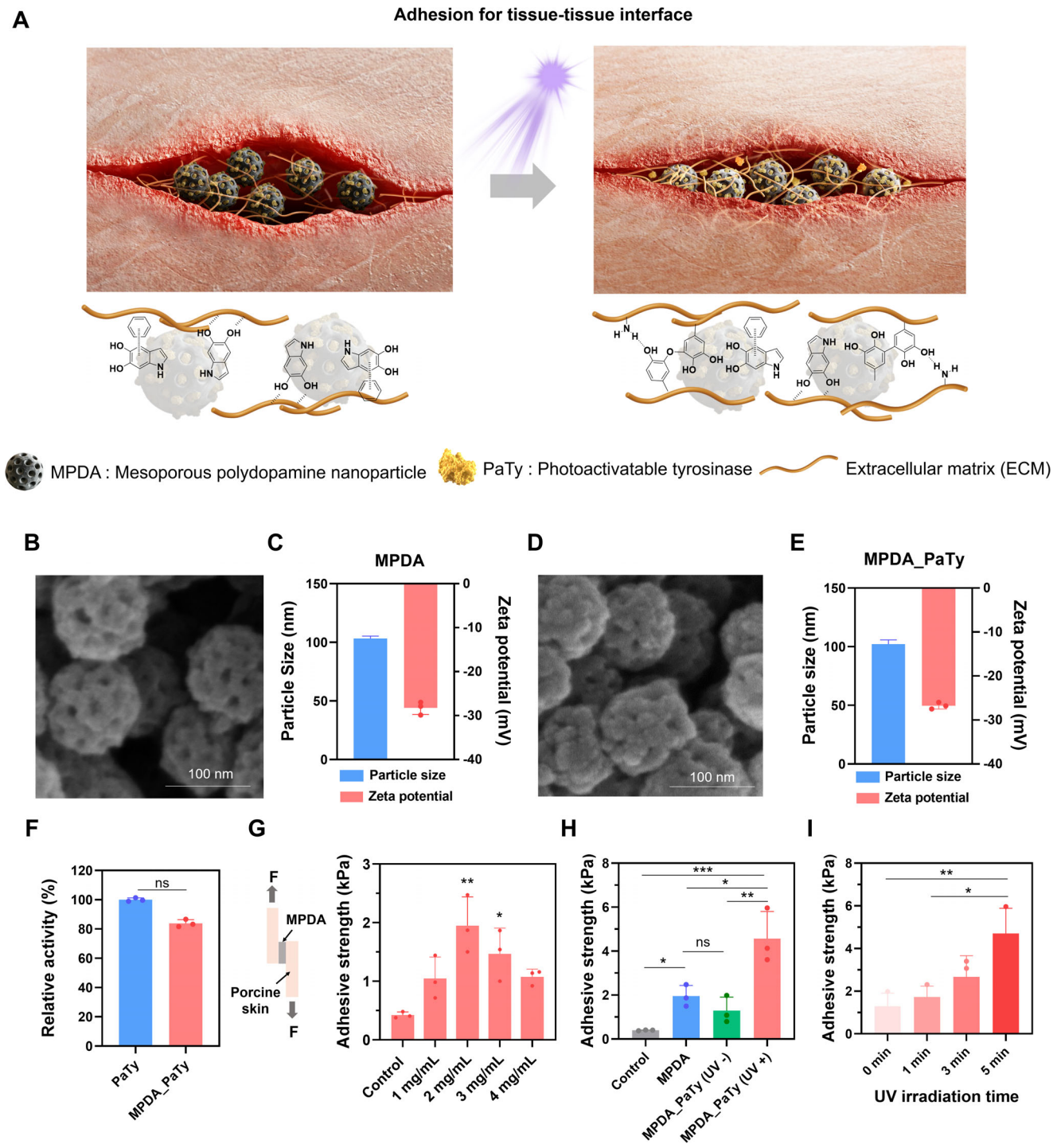

Figure 2. Photoactivatable tyrosinase-loaded mesoporous polydopamine nanoparticle (MPDA_PaTy) was prepared as a light (395 nm, 50 mW cm^{-2})-triggered tissue adhesive. A) Schematic illustration of tissue interfaces with MPDA_PaTy before (left) and after (right) activation. B) SEM image of MPDA was observed. C) The mean size and zeta potential of MPDA were measured. D) SEM image of MPDA_PaTy was observed. E) The mean size and zeta potential of MPDA_PaTy were measured. F) Relative activities of PaTy and MPDA_PaTy were measured. Shear strength was measured for G) MPDA at different concentrations, H) MPDA_PaTy with and without activation, and I) time-dependent activation on porcine skin. Data are presented as the mean ± SD, n = 3. Statistical significance was determined by one-way analysis of variance (ANOVA) with Tukey's post-hoc test. ns, not significant; $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$.

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catechol groups (Figure S6, Supporting information). Peaks at 2945, 1103, and 856 cm^{-1} appeared with MPDA_PaTy, corresponding to C—O—C bonds and peaks from diphenyl or diphenyl ether resulting from the phenoxylic reaction.[19] The increase in phenolic groups and phenoxylic reaction-related peaks corresponds to the enhanced adhesive strength after MPDA_PaTy activation, indicating that the increase in adhesive strength is induced by the crosslinking resulting from PaTy activation.

2.3. Fabrication and Characterization of Stretchable Hydrogel

Furthermore, to demonstrate the synergistic adhesive effect of MPDA_PaTy at the hydrogel-tissue interface, we designed a stretchable hydrogel composed of phenol-functionalized alginate and polyacrylamide (**Figure 3**). Phenol conjugation on the alginate backbone was confirmed with ^{1}H NMR (6.8 and 7.2 ppm; degree of substiton 41.42%, Figure S7A, Supporting information) and FT-IR spectrum (1516 and 1266 cm^{-1} indicate the C═C and C—O bonds of the aromatic ring of tyramine) (Figure S7B). Hydrogel formation from alginate and polyacrylamide was confirmed by rheological analysis, which showed a storage modulus (G′) greater than the loss modulus (G″) (Figure 3B,C).

The rheological properties of the hydrogel with MPDA_PaTy were evaluated to assess the molecular interactions between nanoparticles and the polymer network. Frequency sweep tests revealed that G′ remained consistently higher than G″ in all samples, indicating predominantly elastic behavior (Figure 3B). Notably, both the hydrogel with MPDA and MPDA_PaTy exhibited higher G′ values than the control group, reflecting enhanced intermolecular interactions within the polymer network due to the nanoparticles and PaTy.[28] The molecular interaction between MPDA and the hydrogel increased polymer chain entanglement and network density. Furthermore, following enzymatic crosslinking, activated MPDA_PaTy showed a greater increase in G′ than MPDA alone.

2.4. Evaluation of Hydrogel-Tissue Adhesion by MPDA_PaTy

Activation of MPDA_PaTy enhanced adhesion at the hydrogel-tissue interface (Figure 3D). Adhesive performance of MPDA_PaTy with hydrogel by conducting three standard mechanical tests to measure shear strength (by lap-shear test, ASTM F2255), interfacial toughness (by 180° peel test, ASTM F2256), and wound closure strength (by ASTM F2458-05) (Figure 3E–G).[29–31]

Hydrogel with MPDA showed increased shear strength of 6.7 ± 1.0 kPa, interfacial toughness of 15.4 ± 0.9 J m^{-2}, and wound closure strength of 0.9 ± 0.01 N, compared to the control hydrogel, which exhibited 2.0 ± 0.4 kPa, 9.4 ± 1.9 J m^{-2}, and 0.3 ± 0.1 N, respectively. The MPDA_PaTy (UV+) group further enhanced these properties, achieving 17.9 ± 3.4 kPa, 26.6 ± 0.5 J m^{-2}, and 1.2 ± 0.2 N. In contrast, MPDA_PaTy (UV-) showed a shear strength of 4.6 ± 1.8 kPa, interfacial toughness of 11.1 ± 3.5 J m^{-2}, and wound closure strength of 0.7 ± 0.2 N. The hydrogel with MPDA_PaTy (UV+) also showed adhesion to various organs (Figure S8, Supporting information). Moreover, adhesive strength was maintained up to 150% swelling (Figure S9A, Supporting information), exceeding the expected exudate-induced swelling over a 10-day healing period and thereby indicating sufficient adhesive strength under physiological conditions.[32,33] In addition, the tunable adhesive strength at the hydrogel–tissue interface was demonstrated by adjusting the light irradiation applied to MPDA_PaTy (Figure S9B, Supporting information). Furthermore, using a simple photomask, we generated patterned hydrogel adhesion, demonstrating that adhesion can be selectively enhanced at desired regions by irradiating specific areas (Figure S10, Supporting information).

2.5. Mechanical Wound Closure by Pre-Stretched Hydrogel

The hydrogel was designed for mechanical contraction-induced wound closure by applying it in a pre-stretched form, thereby facilitating effective wound healing. (**Figure 4**). Hydrogel under a stretched ratio ($\lambda_{\text{pre-stretch}}$) of up to 1.3 showed successful contraction to its original size (Figure 4B). The experimental data were compared with the theoretical ratio, assuming that the hydrogel successfully contracted back to its original size. The pre-stretched hydrogel can adhere through on-demand photo-activation, allowing precise control over the timing of adhesion. Upon activation of PaTy, the robust adhesion resists the contraction of the hydrogel alone, and instead, it contracts simultaneously with the tissue and wound (Figure 4C).

Stress and displacement modeling for the wound and hydrogel system were analyzed using 2D axisymmetric finite-element models based on commercially available software (COMSOL Multiphysics). As the stretched hydrogel contracts, it generates a contraction force that creates stress concentration near the wound site, promoting wound closure and enhancing wound healing. We observed that as the hydrogel contracts and displacement increases, the tissue in contact with the hydrogel also contracts, which further alters the displacement. As the displacement magnitude increases, the tissue near the wound sites contracts, reducing the wound diameter (Figure 4D). Due to the native tension in mouse skin, the wound initially enlarges ($\lambda_{\text{initial-enlargement}} = 1.035$) immediately after its generation.[31,34,35] By changing the pre-stretched ratio of the hydrogel (from 1 to 1.3), the contraction ratio was calculated. The contraction appeared when a pre-stretched ratio larger than $\lambda_{\text{pre-strech}} = 1.05$ was applied (Figure 4E). In the $\lambda_{\text{pre-strech}} = 1.3$ pre-stretched system, the calculated radius showed the most significant decrease by 11.2% (Figure 4F). Moreover, wound contraction was observed in a 6 mm circular wound on ex vivo mouse skin to compare with the modeling data (Figure 4C). The $\lambda_{\text{pre-strech}} = 1.3$ pre-stretched hydrogel adhered with activated MPDA_PaTy showed a 12% decrease in the average radial change of the ex vivo wound.

2.6. In Vitro Biocompatibility, UV Protection Ability

Before applying MPDA_PaTy and hydrogel on in vivo models, we demonstrated the in vitro biocompatibility and UV protection capabilities of our system. The biocompatibility of MPDA, MPDA_PaTy, and hydrogel was evaluated using human dermal fibroblasts (HDF). The HDFs maintain viability after 24 h in all groups, showing more than 97% (Figures S11 and S12, Supporting information). Moreover, MPDA_PaTy and the hydrogel were

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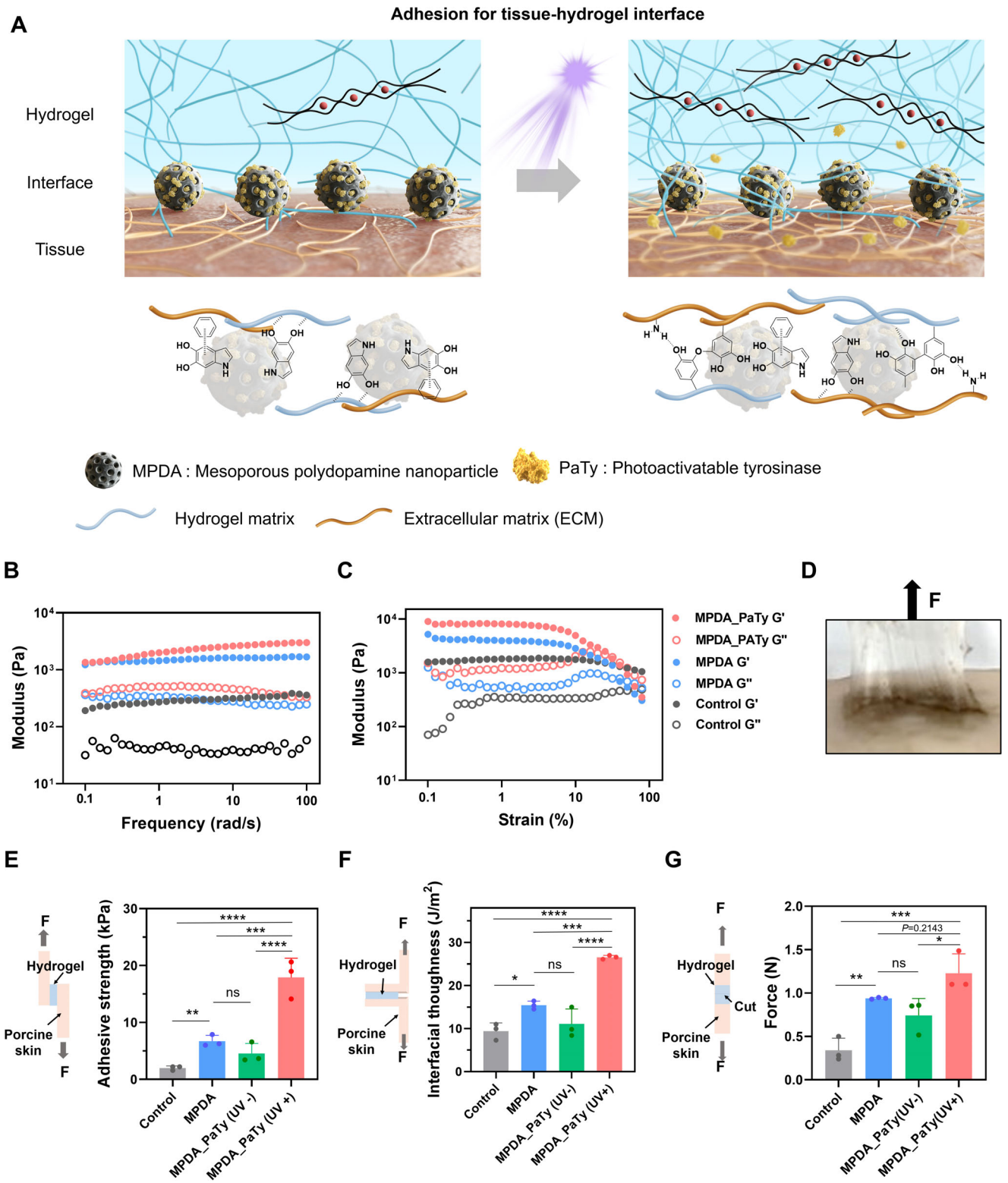

Figure 3. Light (395 nm, 50 mW cm^{-2})-triggered adhesion of MPDA_PaTy was observed at the tissue–hydrogel interfaces. A) Schematic illustration of tissue and hydrogel interfaces with MPDA_PaTy before (left) and after (right) activation. B) Frequency sweep and C) amplitude sweep of the stretchable hydrogel were observed with MPDA and activated MPDA_PaTy. D) Optical image of 90° peeling tests of hydrogels on porcine skin. E) Shear strength, F) Interfacial toughness, and G) wound closure strength of the hydrogel with MPDA_PaTy were measured on porcine skin. Data are presented as the mean ± SD, n = 3. Statistical significance was determined by one-way analysis of variance (ANOVA) with Tukey's post-hoc test. ns, not significant; $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$; $^{****}P < 0.0001$.

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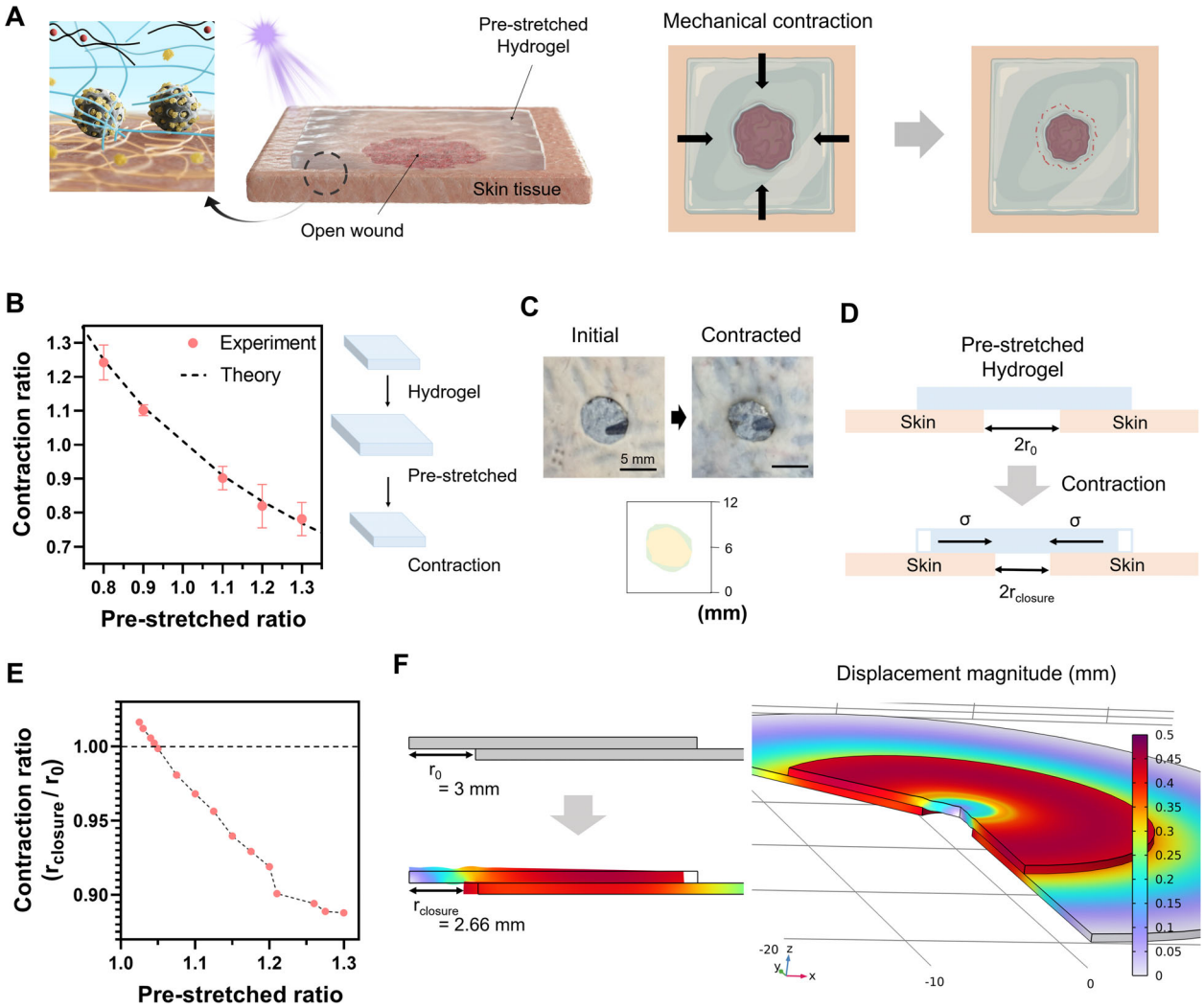

Figure 4. A) Schematic illustration of the application of a pre-stretched hydrogel with reinforced adhesion by activated MPDA_PaTy and mechanical modulation of the skin wound. The two wounded skin figures on the right were created with Biorender.com. B) Theoretical and experimental values of contraction by pre-stretched hydrogels with varying stretch ratios. Data are presented as the mean ± SD, n = 3. C) Representative image of ex vivo mouse skin showing mechanical modulation by a hydrogel fixed with MPDA_PaTy. D) Finite-element analysis setup for mouse skin with wound and hydrogel. E) Results of finite-element analyses showing the wound closure ratio as a function of pre-stretched ratio. The dotted line represents the initial wound size post-creation. F) Representative simulation at a pre-stretched ratio of 1.3. Left: 2D cross-sections showing the skin and hydrogel pre- and post-contraction, with black lines marking their initial positions. Right: 3D visualization reconstructed from 2D axisymmetric finite-element models.

implanted in the subcutaneous region to evaluate biocompatibility and degradation (Figures S13 and S14, Supporting information). Negligible toxicity was indicated by minimal immune cell infiltration in Hematoxylin and Eosin (H&E)-stained tissue and by no significant TNF-α expression in immunohistochemistry (IHC) analysis.

UV protection properties of MPDA were evaluated by calculating the Sun Protection Factor (SPF), critical wavelength, and UVA/UVB ratio. MPDA showed an SPF of 1.12, indicating that MPDA has a 22.4% UV protection capability (Figure S15, Supporting information). Furthermore, the radical scavenging effect was evaluated using 2,2- diphenyl-1-picrylhydrazyl (DPPH) and 2′,7′-dichlorodihydrofluorescein diacetate (DCFDA) assay (Figure S16, Supporting information). MPDA_PaTy reduced DPPH radicals, resulting in a 25% decrease in absorbance compared to the control DPPH group. The DCFDA assay with HDF showed that neither MPDA nor MPDA_PaTy caused an increase in ROS, and both protected HDF from UV-induced ROS production, corresponding to the UV protection ability calculated.

2.7. MPDA_PaTy-Activated On-Demand Adhesion for Incision Wound Closure In Vivo

We evaluated the on-demand adhesion performance of MPDA_PaTy using two different in vivo wound models, one involving a linear incision and the other a circular wound (**Figures 5** and **6**). A mouse model with dorsal incised wounds was selected to confirm the synergistic tissue adhesion effect of

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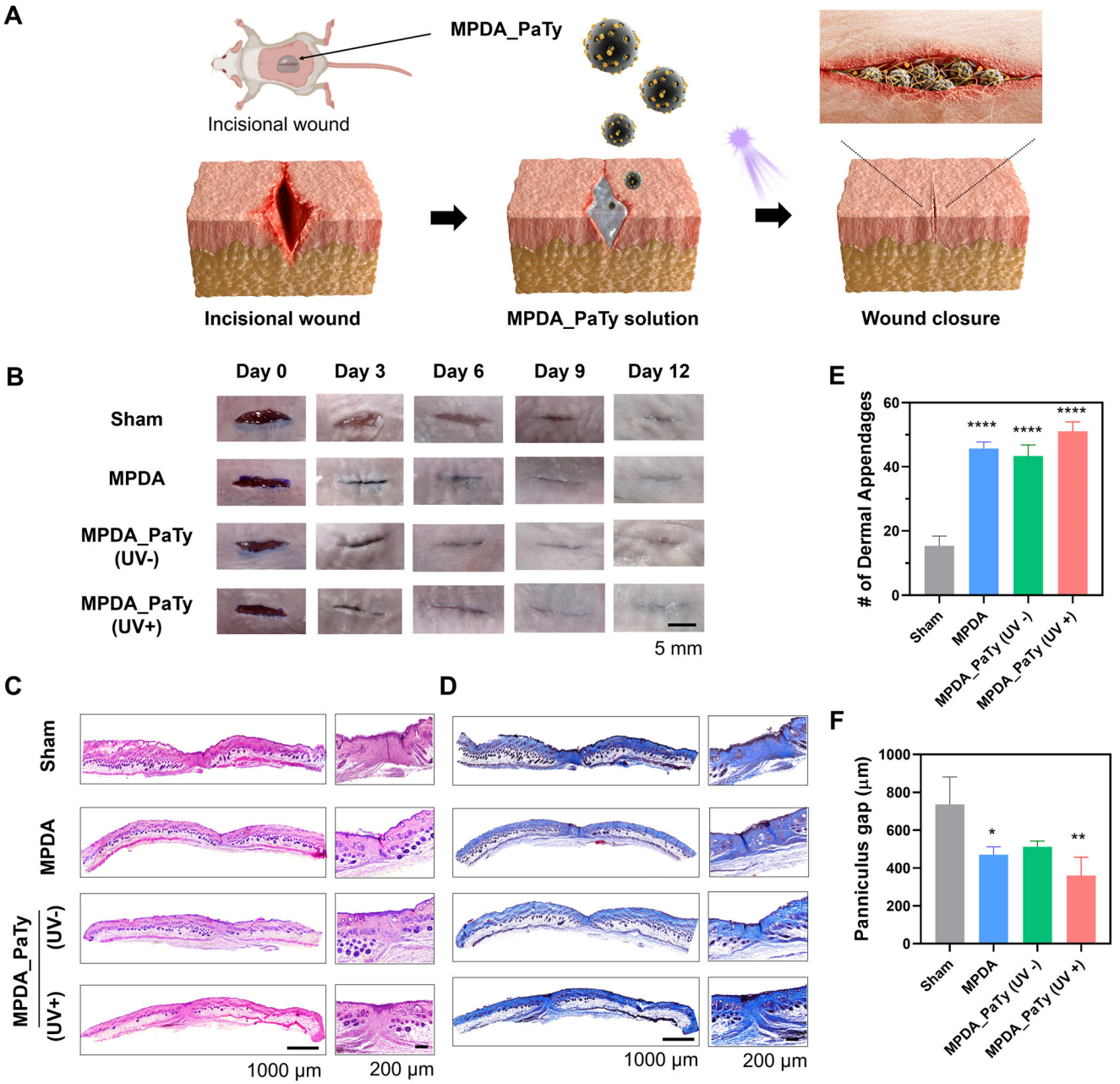

Figure 5. A) Schematic illustration of light (395 nm, 50 mW cm^{-2})-activated MPDA_PaTy application to an incisional wound. The mouse figure was created with Biorender.com. B) Representative images showing the progression of wound regeneration in the control, MPDA, MPDA_PaTy (UV-), and MPDA_PaTy (UV+) groups. Representative cross-sectional histologic C) H&E and (D) MTC images of skin tissue harvested from mice. E) Dermal appendages and F) panniculus gap were compared, evaluating H&E and MTC-stained tissue. Data are presented as the mean ± SD, n = 3. Statistical significance was determined by one-way analysis of variance (ANOVA) with Tukey's post-hoc test. $^{*}P < 0.05$; $^{**}P < 0.01$; $^{****}P < 0.0001$.

MPDA_PaTy (Figure 5A). After making a 1 cm incision, the dorsal skin was treated with different experimental groups: control, only UV (UV+), MPDA, MPDA_PaTy (UV-), and MPDA_PaTy (UV+). Wound images of the incised dorsal skin were observed on days 0, 3, 6, 9, and 12 in each group (Figure 5B; Figure S17, Supporting information). The MPDA_PaTy (UV+) group exhibited the fastest tissue closure, followed by the MPDA and MPDA_PaTy (UV-) groups sequentially.

Histological analysis was performed using H&E staining and Masson's trichrome (MTC) staining on wound samples on day 12 (Figure 5C,D). The MPDA_PaTy (UV+) group exhibited the shortest panniculus gap and the highest number of skin appendages compared to the other groups (Figure 5E,F). This group showed a panniculus gap of 359.3 ± 97.9 μm and 51.0 ±3.0 skin appendages, while the MPDA group showed a panniculus gap of 469.9 ± 41.5 μm and 45.7 ± 2.1 skin appendages. The

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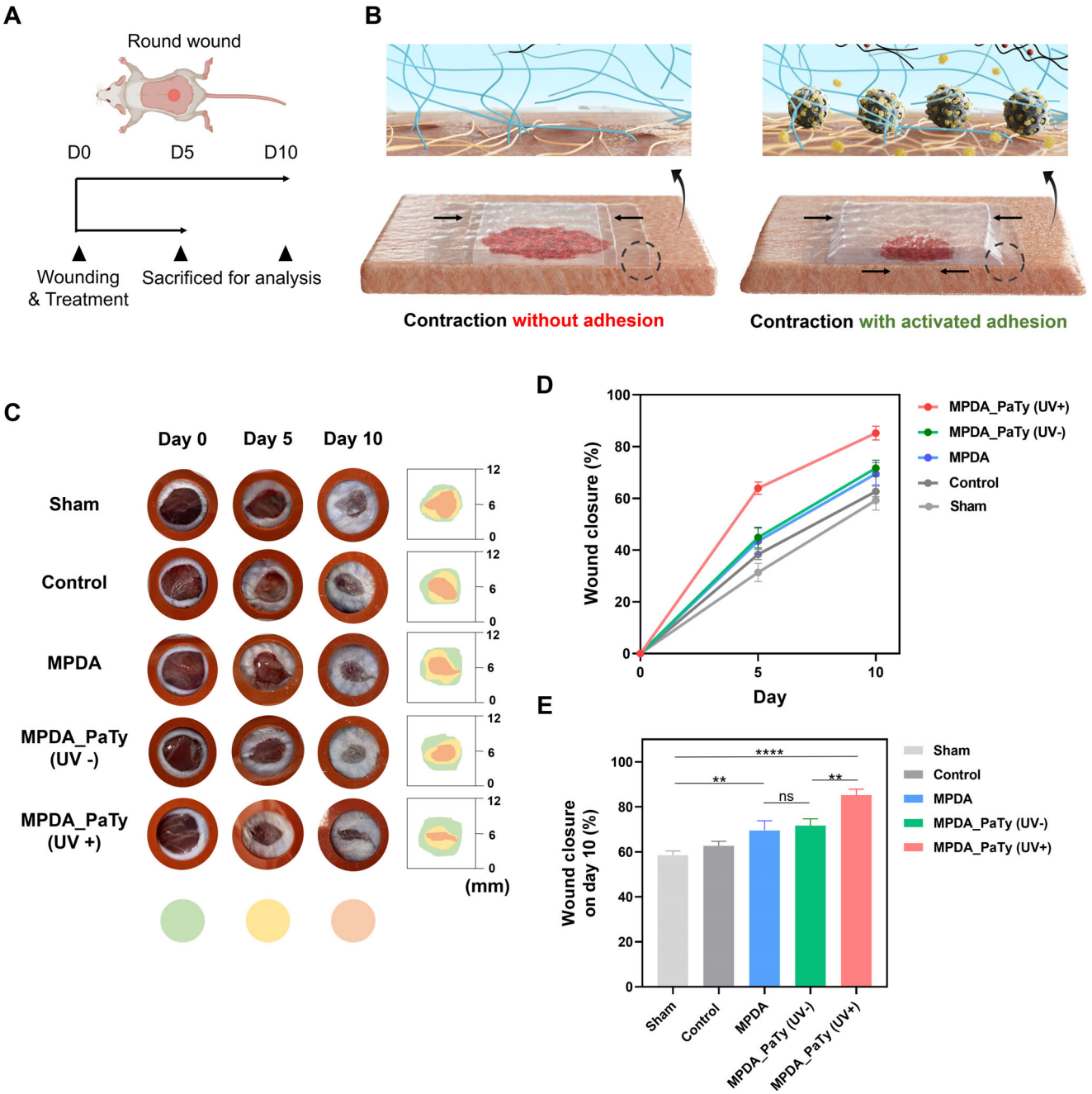

Figure 6. A) Schematic illustration of the experimental plan for the round wound model in mice. The mouse figure was created with Biorender.com. B) Schematic illustration of MPDA_PaTy application to pre-stretched hydrogel without (left) and with light (395 nm, 50 mW cm^{-2})-activated MPDA_PaTy right). C) Representative images showing the progression of wound regeneration in the sham, control, MPDA, MPDA_PaTy (UV-), and MPDA_PaTy (UV+) group. D) Time-course analysis of wound closure percentage over 10 days, and D) comparison of wound closure between groups on day 10. Data are presented as the mean ± SD, n = 6. Statistical significance was determined by one-way analysis of variance (ANOVA) with Tukey's post-hoc test. ns, not significant; $^{**}P < 0.01$; $^{****}P < 0.0001$.

MPDA_PaTy (UV-) group showed a panniculus gap of 512.1 ± 30.4 µm and 43.3 ± 3.5 skin appendages. Meanwhile, the control group displayed a panniculus gap of 735.4 ± 145.6 µm and 15.3 ± 3.1 skin appendages (Figure 5E,F). These results suggest that MPDA can act as a tissue adhesive, while activated PaTy further enhances wound closure through ECM crosslinking.

2.8. In Vivo Round Wound Closure via Mechanical Contraction by MPDA_PaTy-Adhered Hydrogel

The circular wound model was chosen to demonstrate the synergistic and tunable adhesion properties of MPDA_PaTy for the controllable and practical application of a hydrogel patch

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(Figure 6A,B). After creating a wound with a 6 mm radius, the dorsal skin was treated with different experimental groups: sham, control, a hydrogel with UV irradiation (UV+), a hydrogel with MPDA, a hydrogel with MPDA_PaTy (UV-), and a hydrogel with MPDA_PaTy (UV+). All hydrogels are pre-stretched isotropically above the wound with a ratio of $\lambda_{\text{pre-strech}} = 1.3$. Figure 6C and Figure S18 (Supporting information) show wound images and wound size changes on days 0, 5, and 10 for the six groups treated as described above. The MPDA_PaTy (UV+) group showed significantly accelerated wound healing, with 63.9 ± 2.4% closure rates on day 5 and 85.2 ± 2.7% on day 10 (Figure 6D,E). While MPDA_PaTy activation increased adhesion between the tissue and hydrogel, allowing it to withstand contraction forces, other groups with lower adhesive strength exhibited hydrogel contraction without wound contraction, resulting in less effective wound closure. Therefore, MPDA_PaTy (UV-) achieved closure rates of 44.8 ± 4.0% on day 5 and 71.6 ± 3.1% on day 10, while MPDA alone showed rates of 43.5 ± 5.0% on day 5 and 69.5 ± 4.3% on day 10. The control group exhibited closure rates of 38.3 ± 2.0% on day 5 and 62.7 ± 2.1% on day 10, and the sham group showed the rates at 31.4 ± 3.5% on day 5 and 59.2 ± 3.8% on day 10.

Histological analysis was performed using H&E and MTC staining (**Figure 7A–D**; Figure S18, Supporting information). The hydrogel with the MPDA_PaTy (UV+) group exhibited the most significant granulation thickness, measuring 452.5 ± 87.8 µm on day 5 and 524.7 ± 88.3 µm on day 10, compared to the other groups. The hydrogel with the MPDA_PaTy (UV-) group exhibited granulation thicknesses of 209.9 ± 42.2 µm on day 5 and 355.7 ± 50.9 µm on day 10. The hydrogel with the MPDA group showed thicknesses of 310.2 ± 85.0 µm on day 5 and 419.0 ± 86.9 µm on day 10. The control group exhibited thicknesses of 107.0 ± 62.3 µm on day 5 and 265.8 ± 53.8 µm on day 10. The sham group showed granulation thicknesses of 125.1 ± 45.8 µm on day 5 and 141.3 ± 41.1 µm on day 10 (Figure 7E,G).

The hydrogel with the MPDA_PaTy (UV+) group exhibited a re-epithelialization rate of 73.7 ± 14.2% on day 5 and 91.9 ± 11.9% on day 10. The MPDA_PaTy (UV-) group showed rates of 46.0 ± 16.8% on day 5 and 56.9 ± 21.2% on day 10. The MPDA group exhibited rates of 67.5 ± 11.8% on day 5 and 70.7 ± 12.6% on day 10, while the control group showed rates of 33.1 ± 10.2% on day 5 and 47.1 ± 8.9% on day 10. The sham group exhibited the lowest rates of 30.4 ± 12.2% on day 5 and 39.9 ± 7.3% on day 10 (Figure 7F,H).

Potential activation of angiogenesis is also a key indicator of wound healing quality. Angiogenesis is a critical determinant of wound healing quality, as it governs the delivery of oxygen and nutrients to regenerating tissue.[36] Immunostaining images for ***α***-smooth muscle actin (***α***-SMA) and CD31 on day 10 showed that the gel with the MPDA_PaTy (UV+) group had the highest number of new vessels at the wound site compared to all other groups (Figure S19, Supporting information). Specifically, the MPDA_PaTy (UV+) group showed a CD31 vessel density of 3.3 ± 0.2 and 10 ± 1 ***α***-SMA vessels. The MPDA_PaTy (UV-) group showed a CD31 vessel density of 2.5 ± 0.3 and 5.7 ± 0.6 ***α***-SMA vessels. The MPDA group showed a CD31 vessel density of 2.6 ± 0.1 and 6 ± 1 ***α***-SMA vessels. The control group showed a CD31 vessel density of 1.9 ± 0.3 and 3.3 ± 0.6 ***α***-SMA vessels, while the control group exhibited the lowest values, with a CD31 vessel density of 1.0 ± 0.2 and 2.3 ± 0.6 ***α***-SMA vessels.

3. Discussion

Most tissue adhesive patches or glues, including clinically established products such as fibrin glue and cyanoacrylate [17], have been designed for specific conditions with fixed adhesive forces, thereby lacking tunability. The synergistic MPDA_PaTy system was designed to enable tunable, on-demand enhancement of adhesive strength by controlling the duration and intensity of UV irradiation.

While some tunable tissue or cell adhesion systems have been reported [37–39], they require pre-designed scaffold components, resulting in fixed or programmed adhesive strengths. This is primarily because their tunability is achieved through chemical or structural modifications or controlled component ratios. Although some adhesion can be altered by temperature or pH,[40–41], their adhesion cannot be controlled spatiotemporally. Moreover, temperature- and pH-induced systems mainly depend on body temperature and pH. Although some light-responsive bioadhesive systems have reported higher absolute adhesive strengths (e.g., >500kPa[42] and 30–40kPa[43]), these systems typically require pre-functionalized synthetic scaffolds with built-in photoresponsive chemistries, limiting their direct applicability to diverse tissue environments. Our MPDA_PaTy system addresses these challenges using a photo-activatable tyrosinase-based adhesive platform that enables on-demand control and broad applicability in tissue and hydrogel systems. This system introduces a tunable adhesion strategy regulated by light, with tyrosinase enabling broad applicability to diverse phenol-containing interfaces, unlike previous systems.

MPDA was designed to load PaTy and facilitate interaction with tissue interfaces. Tissue proteins containing amine, thiol, or imidazole groups can interact with polydopamine through hydrogen bonding or other non-covalent interactions.[44,45] Moreover, PaTy activation of phenol residues on tissue or hydrogel and the remaining catechol groups on MPDA forms covalent bonds between MPDA and the tissue or hydrogel surfaces. Accordingly, the hydrogel was functionalized with phenol groups to facilitate enzymatic crosslinking by PaTy (Figure S7, Supporting information). Enhanced adhesive properties by MPDA indicate the interaction of non-covalent bonds between the hydrogel and MPDA, which also corresponds to the increase in G′ observed in the rheological evaluation of the hydrogel with MPDA particles (Figure 3E,F).[9,10] Moreover, the increase in G′ with activated MPDA_PaTy and adhesive forces corresponds to the formation of covalent bonds via enzymatic reactions, which can be observed in the FT-IR spectrum of collagen with activated MPDA_PaTy (Figure 2F; Figure S6, Supporting information). This increase in ECM crosslinking can also be demonstrated by the closure of incisional wounds in vivo.

In addition to their adhesive functionality, polydopamine nanoparticles have been reported to exhibit UV-protective properties (Figure S15, Supporting information), which can protect surrounding tissues from UV-related damage.[24,46,47] However, such protection may reduce PaTy activation efficiency. Nevertheless, our studies demonstrate that optimized irradiation conditions ensured sufficient enzyme activation and effective adhesion

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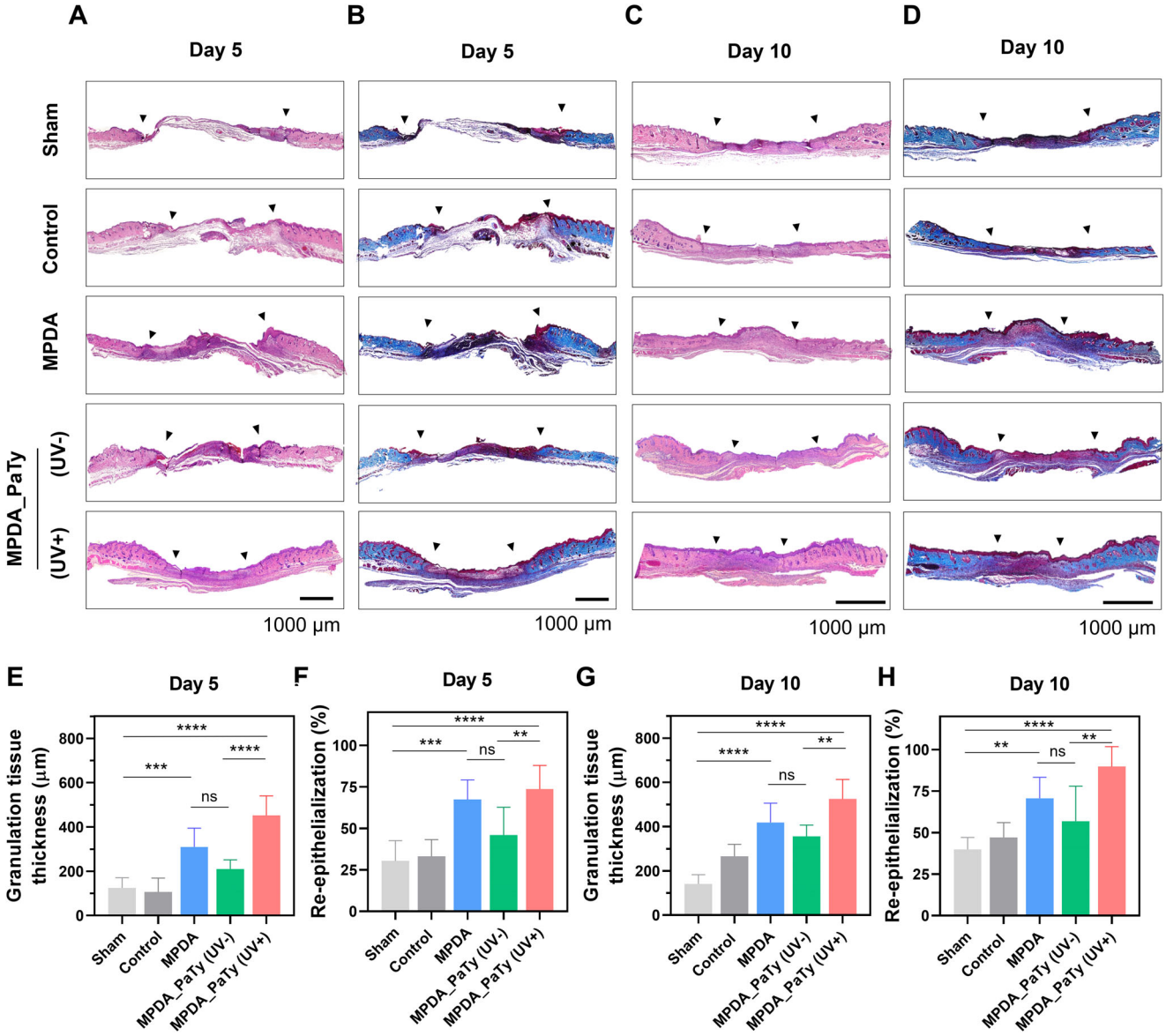

Figure 7. Representative cross-sectional histologic A) H&E and B) MTC images of skin tissue harvested from mice after 5 days of wound generation. Representative cross-sectional histologic C) H&E and D) MTC images of skin tissue harvested from mice after 10 days of wound generation. E) Granulation tissue thickness and F) percentage of re-epithelialization were compared by evaluating H&E and MTC-stained tissue from mice after 5 days of wound generation. G) Granulation tissue thickness and H) percentage of re-epithelialization were compared by evaluating H&E and MTC-stained tissue from mice after 10 days of wound generation. Data are presented as the mean ± SD, n = 6. Statistical significance was determined by one-way analysis of variance (ANOVA) with Tukey's post-hoc test. ns, not significant; $^{**}P < 0.01$; $^{***}P < 0.001$; $^{****}P < 0.0001$.

(Figure 2F–I). Future work may explore MPDA surface modifications or alternative light sources to balance activation efficiency and biological safety.[48]

Biomechanical wound closure has been reported using methods such as strain-programming[31] or closure induced by temperature, pH, and other factors.[49,50] However, most systems have been designed with a fixed stretch ratio or fixed adhesive force. In mechanical wound closure systems, pre-stretched hydrogels must adhere firmly to tissue surfaces while withstanding contraction forces.[51,52] However, strong initial adhesion can hinder the accurate positioning of the patches. Our system enables light-triggered, on-demand adhesion to address the conflicting requirements between re-positionability and adhesive strength. Once activated, the enhanced adhesion provides mechanical support, allowing the hydrogel to resist contraction forces effectively. Since our system can attach via light activation, both the adhesive strength and pre-stretched ratio can be controlled. This controllability enables the regulation of mechanical closure dynamics, as supported by finite-element modeling (Figure 4E,F).

PaTy was initially introduced as a light-responsive enzyme for melanin biosynthesis, with applications in UV protection.[24] In this study, however, PaTy functions as a tunable adhesive module,

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enabling spatiotemporally controlled enzymatic crosslinking in dynamic tissue and hydrogel environments. This shift from passive biochemical protection to active mechanical tissue integration shows the broader potential of photoactivatable enzymes.

While this research focused on two wound models, the MPDA_PaTy system can be used in broader translational cases, such as adhesion between various tissue types or organ surfaces (Figure S8, Supporting information), and could be further optimized with phenol-functionalized hydrogel platforms. These results suggest the potential of MPDA_PaTy as a versatile and clinically relevant platform for dynamic tissue repair, offering precise control, robust adhesion, and compatibility with diverse biomedical environments.

4. Conclusion

In summary, we proposed an on-demand adhesive system based on a photoactivatable enzyme and mesoporous polydopamine nanoparticles. Upon UV irradiation, MPDA_PaTy enabled tunable adhesion through controlled enzymatic activation. This system demonstrated effective adhesion at both tissue-tissue and tissue-hydrogel interfaces. With the in vivo mouse model, MPDA_PaTy facilitated the closure of incision wounds and provided sufficient adhesive strength to support pre-stretched hydrogels, enabling the mechanical contraction of circular wounds. These findings highlight the potential of MPDA_PaTy as a light-activated adhesive platform for a broad range of tissue engineering and regenerative medicine applications.

5. Experimental Section

Materials: Acrylamide, N N N' N'-tetramethylethylenediamine (TEMED), N, N'-methylenebis (acrylamide), sodium alginate, L-tyrosine, ammonium persulfate, dopamine hydrochloride, tyramine hydrochloride, Pluronic F-127, N-hydroxysuccinimide (NHS), and 2,2-diphenyl-1-picrylhydrazyl (DPPH) were purchased from Sigma–Aldrich (St. Louis, MO, USA). N-(3-dimethyl aminopropyl)-N′-ethyl carbodiimide (EDC), Snakeskin dialysis membrane (dialysis membrane), Dulbecco's modified Eagle medium (DMEM), DMEM (high glucose/no glutamine/no phenol red), fetal bovine serum (FBS), penicillin-streptomycin (P/S), LIVE/DEAD Viability/Cytotoxicity Kit, 2′,7′-dichlorodihydrofluorescein diacetate (DCFDA/H2DCFDA), phosphate-buffered saline (PBS) and Pierce BCA Protein Assay Kits were purchased from Thermo Fisher Scientific (UK). *o*-(2-nitrobenzyl)-l-tyrosine hydrochloride was purchased from Accela Chembio Inc. (San Diego, CA, USA). Calcium sulfate dihydrate, ammonium hydroxide, and dimethyl sulfoxide were purchased from Dae Jung (Siheung, Korea). 1,3,5-trimethylbenzene (TMB) was purchased from Tokyo Chemical Industry Co., Ltd (TCI, Tokyo, Japan). Isoflurane was purchased from Hana Pharm (South Korea). Tegaderm was purchased from 3 M (USA). Cell Counting Kit-8 was purchased from DOJIN Laboratories (Kumamoto, Japan). Ni-NTA agarose (nickel-charged resin for His-tag purification) was purchased from Qiagen Korea Ltd. (Seoul, Korea). Rat Tail collagen solution was purchased from Advanced Biomatrix (Carlsbad, CA, USA).

Preparation of Mesoporous Polydopamine Nanoparticle (MPDA): To prepare MPDA, Pluronic F-127 (332 mg) and dopamine hydrochloride (99.6 mg) were dissolved in a mixture of ethanol (10 mL) and distilled water (DW, 10 mL). Then, 199.2 µL of TMB was added. The mixture was dispersed by probe-tip sonication at 30% amplitude for 10 min in an ice bath to form an emulsion and mixed with 796.8 µL of ammonium hydroxide. The raw product particles were obtained by centrifugation at 8000 rpm for 10 min. To remove templates, the particles were washed with a mixture of ethanol and acetone (2:1 v/v) and dispersed by probe-tip sonication at 30% amplitude for 10 min, three times in an ice bath.

The hydrodynamic diameter and zeta potential of the particle were measured by using a Malvern Zetasizer (Nano ZS90, Malvern Panalytical, Malvern, United Kingdom). Morphology of the particle was observed by SEM (JSM-7800F Prime, JEOL Ltd., Japan) and TEM (200 kV; JEM-F200, JEOL Ltd., Japan) at the National Center for Inter-university Research Facilities(NCIRF) at Seoul National University.

Expression and Purification of Photoactivatable Tyrosinase (PaTy): PaTy was synthesized and purified as previously described.[24] Briefly, transformed cells were cultured in 4 mL of LB media containing ampicillin (100 µg mL^{-1}) and chloramphenicol (20 µg mL^{-1}) overnight at 37 °C. These cells were then inoculated into 200 mL of LB media with ampicillin (100 µg mL^{-1}), chloramphenicol (20 µg mL^{-1}), $CuSO_4$ (1 mM), and *o*-nitrobenzyl tyrosine (1 mM). When the optical density at 600 nm reached ≈0.8, protein expression was induced by adding isopropylthio-β-galactoside (IPTG) (0.05 mM) and arabinose (1.33 mM). After incubating for 20 h at 18 °C, cells were harvested by centrifugation at 4000 rpm at 4 °C. The cells were washed with 50 mM Tris-HCl buffer (pH 8.0) and lysed using a sonicator at 30% amplitude for 10 min. The enzyme was then purified using a general His-tag purification as described in previous studies. The final enzyme solution was stored at −20 °C in Tris-HCl buffer (pH 8.0) containing 25% glycerol.

The concentration of the purified PaTy was measured using the Bicinchoninic acid (BCA) Protein Assay Kit (Pierce, Bonn, Germany). A solution containing 200 nM of PaTy in 50 mM Tris-HCl buffer (pH 8.0) was exposed to UV irradiation (395 nm, 50 mW cm^{-2}, 5 min). The intensity of the UV irradiation was measured using the NOVA II from OPHIR (California, USA). Tyrosinase activity was measured using 200 nM PaTy, 10 µM CuSO4, and 2 mM L-tyrosine dissolved in 50 mM Tris-HCl buffer (pH 8.0). The initial rate of the PaTy reaction was determined by measuring the absorbance at 475 nm ($\varepsilon_{475\ \mathrm{dopachrome}}$ = 3600 M^{-1} cm^{-1}) using a UV/VIS spectrophotometer (TECAN Infinite M200 Pro, Mannedorf, Switzerland) at 37 °C.

Preparation of PaTy-Loaded MPDA (MPDA_PaTy): The MPDA was dissolved in PBS (pH 7) containing 5% ethanol. Then, the PaTy solution was added to the particle solution. The mixture was incubated on an orbital shaker at 4 °C overnight. The MPDA_PaTy was obtained by centrifugation at 8000 rpm for 10 min. The MPDA_PaTy was resuspended in PBS containing 5% ethanol and stored at 4 °C for further use. To measure the loading efficiency (LE %), the concentration of loaded PaTy and release profile was calculated using the bicinchoninic acid assay with Pierce BCA Protein Assay Kits.

$$\text{Loading efficiency (LE\%)} = \frac{\text{the final amount of PaTy loaded in MPDA}}{\text{the initial amount of PaTy added}} \times 100\ (\%) \quad (1)$$

MPDA_PaTy was also analyzed by FT-IR spectrometer (Smart iTX, Thermo Fisher Scientific), and each FT-IR spectrum was obtained with 40 scans in the range of 600–4000 cm^{-1}.

Adhesion Properties of MPDA_PaTy: To evaluate the adhesion properties of MPDA and MPDA_PaTy, a lap shear test was conducted using a Universal Testing Machine (UTM, Instron-5543, Instron, USA) equipped with a 50 N load cell. The samples (2 mg mL^{-1}) were applied to a 1.5 cm × 2 cm area of porcine skin (1.5 cm × 2 cm × 3 mm). The area was exposed to UV light for the MPDA_PaTy (UV+) group (395 nm, 50 mW cm^{-2}, 5 min). All tests were performed at a constant tensile speed of 5 mm min^{-1}. Shear strength was calculated by dividing the maximum force by the adhesive area.

Fabrication of the Collagen Hydrogel: Collagen hydrogel was fabricated according to the manufacturing guidelines provided by Advanced Biomatrix, Inc. To prepare the collagen hydrogel, 900 µL of RaT Tail collagen solution (4 mg mL^{-1}) was mixed with 100 µL of a neutralized solution. Subsequently, PBS, MPDA (2 mg mL^{-1}), and MPDA_PaTy (2 mg mL^{-1}) were added to 50 µL of the collagen mixture. For the MPDA_PaTy group, the resulting solution was exposed to UV light (395 nm, 50 mW cm^{-2},

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5 min). FT-IR spectra were obtained using an FT-IR (TENSOR27, BRUKER, Germany).

Preparation of Alginate-Tyramine Conjugate: Tyramine was conjugated to Alginate via EDC/NHS coupling. To prepare Alginate-tyramine conjugate, 1 g of alginate was dissolved in 100 mL of DW. 1.738 g of EDC, 1.3 g of NHS, and 1.5 g of tyramine hydrochloride were added to the solution. The mixture was reacted for 24 h at RT, followed by dialysis for 48 h against 3.5K MWCO membrane (SnakeSkin Dialysis Tubing; 35 mm, Sigma) to remove residuals. After lyophilization, the final product was obtained. The Alginate-tyramine conjugate was analyzed by ^{1}H NMR (300 MHz, D_2O) (AvanceIII HD, BRUKER, Germany) and FT-IR spectrometer (TENSOR27, BRUKER, Germany), and each FT-IR spectrum was obtained with 32 scans in the range of 400–4000 cm^{-1}.

Preparation of the Stretchable Hydrogel: To prepare an Alginate-tyramine and polyacrylamide hydrogel, sodium alginate and synthesized alginate-tyramine conjugate were dissolved in DW in a 1:1 mass ratio to achieve a final concentration of 2.2%. Acrylamide was then added to the solution at 13.5%. The mixture of 16 mL was mixed with 304.8 µL of 0.75 M calcium sulfate dihydrate, 57.4 µL of 2% N, N′-methylenebis (acrylamide), 12.7 µL of TEMED, and 316.4 µL of 6.6% ammonium persulfate. The solution was polymerized to hydrogel in a PDMS mold at room temperature overnight. Then, the fabricated hydrogel was washed in PBS 3 times.

Mechanical Characterization of the Hydrogel: The characterization was conducted with a control group, an MPDA group, a MPDA_PaTy (UV-) group, and a MPDA_PaTy (UV+) group. Different solutions (MPDA, MPDA_PaTy) were initially applied to the porcine skin tissue, and then the hydrogel was placed on top. For the MPDA_PaTy (UV+), UV light exposure (Ultrafire WF-501B, 395 nm, 50 mW cm^{-2}, 5 min) was applied before testing.

To measure shear strength, the hydrogel was applied to a 1.5 cm × 2 cm area of porcine skin (1.5 cm × 10 cm × 3 mm) and tested using a lap shear test with a mechanical testing machine (UTM, Instron-5543, Instron, USA) equipped with a 100 N load cell. All tests were performed at a constant rate of 50 mm min^{-1}.

To measure interfacial toughness, the hydrogel was applied to a 2.5 cm × 2.5 cm area of porcine skin (2.5 cm × 10 cm × 3 mm), and a standard 180-degree peel test was performed using a mechanical testing machine (UTM, Instron-5543, Instron, USA). All tests were performed at a constant rate of 50 mm min^{-1}.

To measure wound closure strength, the hydrogel, with a 3 cm overlap length and a width of 1 cm, was applied to the porcine skin (1.5 cm × 10 cm × 3 mm) and tested using the wound closure strength test with a mechanical testing machine (UTM, Instron-5543, Instron, USA). All tests were performed at a constant rate of 50 mm min^{-1}. Wound closure strength was assessed by measuring the maximum force achieved.

Rheological tests were performed on the hydrogel with MPDA and MPDA_PaTy using a rheometer (ARES, Rheometric Scientific, UK). The hydrogels were first hydrated in a sample and then placed on the base plate. To activate PaTy, UV light (Ultrafire WF-501B, 395 nm, 50 mW cm^{-2}, 5 min) was applied to the MPDA_PaTy group before testing. Strain sweeps data were measured by the storage modulus (G') and loss modulus (G″) at varying shear strains from 0.1% to 100% at a fixed frequency of 1 Hz. Frequency sweeps measured G' and G″ while increasing the frequency from 0.1 to 10 Hz with a fixed shear strain of 10%.

Ex Vivo Wound Closure: Skin specimens were harvested from BALB/c mice, and circular wounds with a diameter of 6 mm were created using a biopsy punch. Pre-stretched hydrogels ($\lambda_{\text{pre-stretch}} = 1.3$) with MPDA_PaTy (UV+) were applied to the wounds. The samples were incubated for 5 min at 37 °C and then frozen in a deep freezer. After the removal of the hydrogel, the wounds were photographed to measure the wound size. The areas of the wound were analyzed using ImageJ software.

Finite-Element Modeling of Wound Contraction: 2D axisymmetric finite element models were developed using COMSOL Multiphysics to simulate mechanical wound contraction by the pre-stretched patch. The skin tissue was modeled as an incompressible Ogden hyperelastic material with coefficients calculated from the fitting of stress-strain curves of the dorsal skin of a BALB/c mouse (µ = 36 kPa, α = 18), incorporating a pre-strain of 3.5% to reflect the native skin tension.[31,34,35] The adhesive patch was modeled as a neo-Hookean material with a shear modulus of 259.7 kPa. Boundary conditions were defined to replicate post-adhesion equilibrium with symmetric deformation about the wound center. The simulations were used to analyze displacement magnitude and to evaluate the wound closure dynamics as a function of initial patch strain. The displacement variables were extracted to assess deformation-induced remodeling in the surrounding tissue.

In Vitro Biocompatibility of MPDA_PaTy and the Hydrogel: Human dermal fibroblasts (HDF) were seeded into 96-well plates at 1×10^4 cells well^{-1} density and cultured at 37 °C with 5% CO_2 for 24 h. The hydrogel was immersed in 10 mL of growth media (DMEM with 10% FBS and 1% P/S) for 24 h at 37 °C to generate liquid extracts from the hydrogel. The following day, MPDA (2 mg mL^{-1}), MPDA_PaTy (2 mg mL^{-1}), and hydrogel extracts were added to each well. After incubating for 24 h, a Live/Dead assay was performed using the LIVE/DEAD cell viability kit (Thermo Fisher Scientific).

Furthermore, the CCK-8 assay was performed for up to 3 days using the Cell Counting Kit-8 (DOJIN Laboratories). Add 10 µL of CCK-8 solution to the cell medium. Plates were incubated at 37 °C in the dark for 2 h. The proliferation ability was determined by measuring the absorbance at 450 nm using a UV/VIS spectrophotometer (TECAN Infinite M200 Pro, Männedorf, Switzerland).

In Vivo Biocompatibility and Biodegradation of MPDA_PaTy and the Hydrogel: MPDA_PaTy (2 mg mL^{-1}, 100 µL) and the hydrogel (1.0 cm × 1.0 cm) were implanted in the subcutaneous region to evaluate both biocompatibility and biodegradation. All animal studies and experimental procedures were approved by the Institutional Animal Care and Use Committee of Seoul National University (Registration No. SNU-240730-7). Normal 7-week-old BALB/c female mice (n = 3 per group) were obtained from Orient-Bio (Republic of Korea). Ten days after implantation, skin tissues were collected, fixed in 4% paraformaldehyde (PFA), and embedded in paraffin for histological analysis. Hematoxylin and eosin (H&E) staining was performed according to standard protocols.

For immunohistochemistry staining, paraffin-embedded skin tissues were incubated with 5% normal goat serum (NGS) for 1 h at room temperature to block nonspecific binding. Subsequently, the samples were stained overnight at 4 °C with the primary antibody, rabbit anti-mouse TNF-α. After PBS washing, the samples were incubated with secondary antibody, Alexa Fluor 488 goat anti-rabbit IgG. Finally, the slides were mounted with DAPI and analyzed using an inverted optical microscope (Eclipse Ti2, Nikon, Japan). Fluorescence intensities were quantified using ImageJ software.

UV Protection Ability of MPDA: Sun protection factor (SPF), critical wavelength (λ_c), and UVA/UVB ratio were calculated as described in previous studies,[19][24]

SPF was calculated using the following equations:

$$SPF_{spectrophotometic} = CF \times \sum_{290}^{320} EE(\lambda)\, I(\lambda)\, \text{Abs}(\lambda) \tag{2}$$

In these calculations, CF represents the correction factor (= 10), EE (λ) was the erythemal effect spectrum, I (λ) was the solar intensity spectrum, and Abs (λ) was the absorbance of the solution measured by UV/vis spectroscopy (TECAN Infinite M200 Pro, Mannedorf, ¨Switzerland). The EE and I values were calculated following the reference method.[53]

The critical wavelength value was calculated using the following equations:

$$\int_{290}^{\lambda c} A(\lambda)\, d\lambda = 0.9 \int_{290}^{400} A(\lambda)\, d\lambda \tag{3}$$

The UVA/UVB ratio was calculated using the following equations:

$$\frac{UVA}{UVB} ratio = \frac{\int_{290}^{400} A(\lambda)\, d\lambda / \int_{290}^{400} d\lambda}{\int_{290}^{320} A(\lambda)\, d\lambda / \int_{290}^{320} d\lambda} \tag{4}$$

where A (λ) represents the average absorbance at each wavelength.[54]

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Radical Scavenging Effect of MPDA: DPPH assay was performed using a standard procedure to observe the radical scavenging effect of MPDA and MPDA_PaTy. A working solution of 0.2 mM DPPH in methanol was prepared. 0.2 mM DPPH was made for the working solution in a methanol solution. The DPPH solution of 180 µL was mixed with 20 µL of either MPDA (2 mg mL^{-1}) or MPDA_PaTy (2 mg mL^{-1}). The solution was incubated at 37 °C in the dark for 30 min. The radical scavenging activity was determined by measuring the absorbance at 517 nm using a UV/VIS spectrophotometer (TECAN Infinite M200 Pro, Männedorf, Switzerland).

DCFDA assay was performed following the manufacturer's guidelines to observe the reactive oxygen species (ROS) scavenging effect of MPDA and MPDA_PaTy. A 20 mM DCFDA stock solution was prepared in DMSO and diluted with PBS to 20 µM. The culture media for the DCFDA assay were prepared by supplementing DMEM (high glucose, without glutamine and phenol red) with 10% FBS, 1% penicillin-streptomycin (P/S), and 2 mM GlutaMAX. HDF were seeded into 96-well plates at a density of 3.0×10^4 cells per well at 37 °C with 5% CO_2. The following day, after the cells were washed with PBS, 100 µL of DCFDA solution (20 µM) was treated. The plates were incubated at 37 °C in the dark for 45 min. After incubation, the DCFDA solution was removed and washed with PBS. 100 µL of MPDA (2 mg mL^{-1}) or MPDA_PaTy (2 mg mL^{-1}) was added to each well and incubated for 1 h. To induce ROS, the cells were exposed to UV irradiation (395 nm, 50 mW cm^{-2}, 5 min). After 30 min, the fluorescence intensity of the 2′,7′-dichlorofluorescein (DCF) was measured using a UV/vis spectrophotometer (TECAN Infinite M200 Pro, Mannedorf, Switzerland) at an excitation wavelength of 488 nm and emission wavelength of 535 nm.

In Vivo Incisional Wound Closure: The tissue adhesive effect of MPDA_PaTy was verified on the in vivo incisional wound closure model. All animal studies and experimental procedures were approved by the Institutional Animal Care and Use Committee at Seoul National University (Registration No. SNU-240730-7). Normal 8-week-old BALB/c mice (male; n = 3 per group) were obtained from Orient-Bio (Republic of Korea). The mice were anesthetized and shaved, followed by creating a 1 cm incisional wound on the dorsal skin. 25 µL of the sample (2 mg mL^{-1}) was treated to the incised dorsal skin. The test was conducted with a sham group, only UV (UV+) group, a MPDA group, a MPDA_PaTy (UV-) group, and a MPDA_PaTy group (UV+) group. For the MPDA_PaTy (UV+) group, the mice were treated with UV (Ultrafire WF-501B, 395 nm, 50 mW cm^{-2}, 5 min). UV irradiation was performed 5 cm above the wound spot. Tegaderm was used to protect the skin tissue. Each incised dorsal skin was observed at five time points of day 0, day 3, day 6, day 9, and day 12. After the experiment, mice were euthanized using excess CO_2, and the dorsal skin at the wound site was collected.

In Vivo Wound Closure by Mechanical Contraction of Hydrogel: The wound mechanical closure effect of pre-stretched hydrogel using a tunable adhesion system was evaluated in a round wound-induced mouse model. All animal studies and experimental procedures were approved by the Institutional Animal Care and Use Committee at Seoul National University (Registration No. SNU-240730-7). Normal 8-week-old BALB/c mice (male; n = 6 per group) were obtained from Orient-Bio (Republic of Korea). The mice were anesthetized and shaved, and then a 6 mm round wound was created on the dorsal skin using a biopsy punch. The experimental groups were divided as follows: (1) sham, (2) control (hydrogel with PBS), (3) only UV (UV+) group, (4) MPDA, (5) MPDA_PaTy (UV-), and (6) MPDA_PaTy (UV+). After spreading a droplet of a nanoparticle solution (50 µL) around the skin wound, the hydrogel (1.0 cm × 1.0 cm) was placed above the solution. All hydrogels were pre-stretched isotropically above the wound with a ratio of $\lambda_{\text{pre-stretch}} = 1.3$. In the MPDA_PaTy (UV+) group, the MPDA_PaTy was exposed to UV (Ultrafire WF-501B, 395 nm, 50 mW cm^{-2}, 5 min). UV irradiation was performed 5 cm above the wound spot, irradiating the tissue–hydrogel interface. Each skin wound image was captured on day 0, day 3, day 6, day 9, and day 12, and wound healing rates were calculated using ImageJ software. On days 6 and 12, mice were euthanized using excess CO_2, and dorsal skin at the wound site was collected for further analysis.

Histological Analysis: The skin tissue was fixed with 4 % paraformaldehyde (PFA) and embedded in paraffin for histological analysis. The skin samples were sectioned at a thickness of 5 µm. Hematoxylin and eosin (H&E) and Masson's trichrome (MTC) staining were performed according to the conventional methods. The slides were observed using an inverted optical microscope (Eclipse Ti2, Nikon, Japan). The Panniculus gap, the number of dermal appendages, granulation tissue thickness, and re-epithelialization were estimated using ImageJ software.

For immunohistochemistry staining, the paraffin-sectioned skin tissues were incubated with 5% normal goat serum (NGS) for blocking for 1 h at room temperature. Subsequently, the samples were then stained overnight at 4 °C with primary antibodies, anti-CD31(1:100; Invitrogen; MA1-80069) and anti-α-smooth muscle actin (αSMA; 1:100; abcam; ab5694). After PBS washing, the samples were incubated in secondary antibodies, Alexa 594 goat anti-rat antibody IgG (1:1000; Invitrogen; A48264) and Alexa 488 goat anti-rabbit antibody IgG (1:1000; Invitrogen; A11008). Next, the slides were mounted with DAPI (1:1000; Thermo Fisher Scientific) and analyzed with an inverted optical microscope (Eclipse Ti2, Nikon, Japan). The fluorescence intensities were measured using ImageJ software.

Statistical Analysis: All data ($n \geq 3$) were processed as described in the Experimental Section. Values were presented as mean and standard deviation (mean ± SD). Statistical analyses were performed using GraphPad Prism 8 software (GraphPad Software; USA). Data were analyzed using one-way ANOVA with Tukey's post-hoc test. *P*-values were denoted as follows: ns (not significant), * ($P < 0.05$), ** ($P < 0.01$), *** ($P < 0.001$), **** ($P < 0.0001$).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

enzyme-triggered crosslinking, mesoporous nanoparticles, photo-activation, tunable adhesion, wound contraction

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